

POLYNOMIAL CERTIFICATES FOR THE FIRST TWO PARTIAL QUOTIENTS OF $\sqrt{(\zeta(3) - S_N)/2}$, AND THE THIRD FOR $N \geq 2$

ABSTRACT. Let $S_N = \sum_{j=1}^N j^{-3}$, let $R_N = \zeta(3) - S_N$ and let $g(N) = \sqrt{R_N/2}$. Feldman displays the continued fraction $g(N) = [0; 2N + 1, 4N + 2, \lfloor (12N + 6)/19 \rfloor, \dots]$ and records, as the first of his open problems, that his uniform Euler–Maclaurin remainder does not pin even the first two of these at the smallest index; in particular his claim that $a_1(g)|_{N_k} = Q_{2k+1}$ and $a_2(g)|_{N_k} = 2Q_{2k+1}$ along the Pell indices N_k is left unproved. We prove $a_1(g(N)) = 2N + 1$ and $a_2(g(N)) = 4N + 2$ for every integer $N \geq 1$, which gives that claim at every $k \geq 1$. We also prove $a_3(g(N)) = \lfloor (12N + 6)/19 \rfloor$ for every $N \geq 2$; at $N = 1$ the displayed value 0 is not the third partial quotient of $g(1)$, which is 1. Each floor decision is reduced to the non-negativity of the coefficient vector of an explicit integer polynomial; no value of $\zeta(3)$ is used in any proof. The trailing ellipsis of the display is untouched, so Feldman’s conjecture as a whole remains open.

1. THE STATEMENT

For a real x we write $x = [a_0; a_1, a_2, \dots]$ for its regular continued fraction and $a_i(x)$ for its i -th partial quotient. Throughout,

$$S_N = \sum_{j=1}^N \frac{1}{j^3}, \quad R_N = \zeta(3) - S_N = \sum_{j>N} \frac{1}{j^3}, \quad g(N) = \sqrt{\frac{R_N}{2}}.$$

Since $R_1 = \zeta(3) - 1 < 2$ we have $0 < g(N) < 1$, so $a_0 = 0$ and the index 1 really is the first floor-and-invert. Also $g(N)$ is irrational for every N : $\zeta(3)$ is irrational [2, 3] and S_N is rational, so $R_N \notin \mathbb{Q}$, and a rational $g(N)$ would make $R_N = 2g(N)^2$ rational. Hence the continued fraction of $g(N)$ is infinite and every partial quotient below is unambiguous.

Feldman [1] introduces g and states the following, quoted from the e-print source.

Conjecture (Feldman) ([1]). The formal continued fraction of $g(N) = \sqrt{R_N/2}$ is

$$(1) \quad g(N) = \left[0; 2N + 1, 4N + 2, \left\lfloor \frac{12N + 6}{19} \right\rfloor, \dots \right]$$

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with quasi-polynomial partial quotients of periods $1, 1, 19, \dots$. In particular, at $N = N_k$:

$$(2) \quad a_1(g)|_{N_k} = 2N_k + 1 = Q_{2k+1}, \quad a_2(g)|_{N_k} = 4N_k + 2 = 2Q_{2k+1}.$$

Here P_k are the Pell numbers $0, 1, 2, 5, 12, 29, 70, 169, \dots$, $Q_k = P_k + P_{k-1}$, $M_k = P_{2k+1}$ and $N_k = (Q_{2k+1} - 1)/2 = 3, 20, 119, 696, \dots$, so that $Q_{2k+1}^2 - 2M_k^2 = -1$. Thus (2) is the restriction of the first two entries of (1) to $N = N_k$, and is strictly weaker than (1).

What is open is stated by the author himself, at the first of his open problems, again quoted:

“The derivation is only formal: the uniform Euler–Maclaurin remainder ... does not pin the leading floors at the smallest index $k = 1$ ($M_k = 5$), so even the first two quotients $a_1(g)|_{N_k} = Q_{2k+1}$, $a_2(g)|_{N_k} = 2Q_{2k+1}$ are not proved there. A sharper index-dependent remainder, or a direct treatment of the small cases, would settle it.”

The author’s main theorem, which is proved, concerns $\sqrt{R_{N_k}}$ — a different real number, $\sqrt{2}$ times ours — and only for $k \geq 3$, so it says nothing at $k = 1$.

We prove the following.

Theorem 1. *For every integer $N \geq 1$,*

$$a_1(g(N)) = 2N + 1 \quad \text{and} \quad a_2(g(N)) = 4N + 2.$$

Theorem 2. *For every integer $N \geq 2$,*

$$a_3(g(N)) = \left\lfloor \frac{12N + 6}{19} \right\rfloor,$$

and this fails at $N = 1$: there $a_3(g(1)) = 1$ while $\lfloor 18/19 \rfloor = 0$.

Theorem 1 discharges (2) at every $k \geq 1$, the index $k = 1$ ($N_1 = 3$, $M_1 = 5$) included, which is exactly the statement the author’s open problem names as unproved. Theorem 2 is an additional numeric statement about the third entry of (1), with $N = 1$ excluded. What is *not* settled is recorded in Section 6.

The mechanism is elementary, which is why no value of $\zeta(3)$ appears: we build two short antidifferences of opposite defect sign, obtaining a tail-free two-sided enclosure of R_N valid from $N = 1$; each floor decision then becomes the positivity of a rational function of N whose shifted numerator has non-negative integer coefficients.

2. TWO ANTIDIFFERENCES OF OPPOSITE DEFECT SIGN

Put

$$(3) \quad \psi(x) = \frac{1}{2x^2} + \frac{1}{2x^3} + \frac{1}{4x^4} - \frac{1}{12x^6} = \frac{A(x)}{12x^6}, \quad \psi_3(x) = \psi(x) + \frac{1}{12x^8} = \frac{B(x)}{12x^8},$$

where

(4)

$$A(x) = 6x^4 + 6x^3 + 3x^2 - 1, \quad B(x) = x^2 A(x) + 1 = 6x^6 + 6x^5 + 3x^4 - x^2 + 1.$$

Both vanish at infinity. Define the defects

$$D(j) = \psi(j) - \psi(j+1) - \frac{1}{j^3}, \quad D_3(j) = \psi_3(j) - \psi_3(j+1) - \frac{1}{j^3}.$$

Lemma 3. *As identities of polynomials in j ,*

$$(5) \quad 12j^6(j+1)^6 D(j) = -(2j+1)^3 = -(8j^3 + 12j^2 + 6j + 1),$$

$$(6) \quad 12j^8(j+1)^8 D_3(j) = 18j^5 + 45j^4 + 48j^3 + 27j^2 + 8j + 1.$$

Hence $D(j) < 0$ and $D_3(j) > 0$ for every $j \geq 1$.

Proof. Both are polynomial identities after clearing denominators, and both are verified by expansion. Explicitly, (5) is $A(j)(j+1)^6 - A(j+1)j^6 - 12j^3(j+1)^6 = -(2j+1)^3$, and (6) is $B(j)(j+1)^8 - B(j+1)j^8 - 12j^5(j+1)^8 = 18j^5 + 45j^4 + 48j^3 + 27j^2 + 8j + 1$. The two coefficient vectors $(-1, -6, -12, -8)$ and $(1, 8, 27, 48, 45, 18)$, in ascending powers of j , have all entries of one sign, and the signs are opposite; the denominators $12j^6(j+1)^6$ and $12j^8(j+1)^8$ are positive for $j \geq 1$. As a hand check at $j = 1$: $\psi(1) - \psi(2) - 1 = \frac{7}{6} - \frac{155}{768} - 1 = -\frac{27}{768}$ and $\psi_3(1) - \psi_3(2) - 1 = \frac{49}{1024}$. $\square$

Lemma 4. *For every integer $N \geq 1$,*

$$(7) \quad \psi(N+1) < R_N < \psi_3(N+1),$$

and moreover

$$(8) \quad \psi_3(N+1) - \frac{49}{40N^{10}} < R_N < \psi_3(N+1).$$

Proof. Since $\psi(x) \rightarrow 0$, the series $\sum_{j>N} (\psi(j) - \psi(j+1))$ telescopes exactly to $\psi(N+1)$, whence

$$R_N = \sum_{j>N} \frac{1}{j^3} = \sum_{j>N} (\psi(j) - \psi(j+1)) - \sum_{j>N} D(j) = \psi(N+1) - \sum_{j>N} D(j),$$

and likewise $R_N = \psi_3(N+1) - \sum_{j>N} D_3(j)$. Both tail series converge absolutely, since (5) and (6) give $|D(j)| = O(j^{-9})$ and $D_3(j) = O(j^{-11})$. By Lemma 3 the first tail is negative and the second positive, which is (7); note that the upper bound carries *no* remainder term.

For (8) we bound the second tail. From (6), $D_3(j) \leq \frac{49}{4}j^{-11}$ for every $j \geq 1$: clearing denominators, the claim is that $588(j+1)^8 - 4j^3(18j^5 + 45j^4 + 48j^3 + 27j^2 + 8j + 1)$ has non-negative coefficients, and indeed it equals $516j^8 + 4524j^7 + 16272j^6 + 32820j^5 + 41128j^4 + 32924j^3 + 16464j^2 + 4704j + 588$. Next, $\sum_{j>N} j^{-11} \leq \frac{1}{10}N^{-10}$, because $\frac{1}{10}j^{-10} - \frac{1}{10}(j+1)^{-10} \geq (j+1)^{-11}$ for $j \geq 1$ — which after clearing denominators reads $(j+1)^{11} - j^{11} - 11j^{10} \geq 0$, true coefficientwise — and the left side telescopes. Hence $0 < \sum_{j>N} D_3(j) \leq \frac{49}{4} \cdot \frac{1}{10}N^{-10}$. $\square$

3. THE FIRST TWO QUOTIENTS

Write $V = 1/g(N) = \sqrt{2/R_N}$ and

$$(9) \quad \begin{aligned} \ell(N) &= (2N+1) + \frac{1}{4N+3} = \frac{8N^2+10N+4}{4N+3}, \\ h(N) &= (2N+1) + \frac{1}{4N+2} = \frac{8N^2+8N+3}{4N+2}. \end{aligned}$$

Proof of Theorem 1. Consider the two rational functions

$$I_2(N) = h(N)^2 \psi(N+1) - 2, \quad I_1(N) = 2 - \ell(N)^2 \psi_3(N+1).$$

Clearing denominators, $I_2 = \mathcal{I}_2/(12(N+1)^6(4N+2)^2)$ and $I_1 = \mathcal{I}_1/(12(N+1)^8(4N+3)^2)$ with $\mathcal{I}_1, \mathcal{I}_2 \in \mathbb{Z}[N]$; substituting $N = 1+t$,

$$(10) \quad \mathcal{I}_2(1+t) = 659 + 1380t + 1055t^2 + 342t^3 + 38t^4,$$

$$(11) \quad \begin{aligned} \mathcal{I}_1(1+t) &= 492 + 3448t + 7664t^2 + 8328t^3 \\ &\quad + 5032t^4 + 1728t^5 + 316t^6 + 24t^7. \end{aligned}$$

Every coefficient in (10) and (11) is positive, and both denominators are manifestly positive for $N \geq 1$; therefore $I_1(N) > 0$ and $I_2(N) > 0$ for every integer $N \geq 1$. (As a hand check of (11) at $N = 1$: $2 \cdot 12 \cdot 256 \cdot 49 - 484 \cdot 621 = 301056 - 300564 = 492$. Values of $\mathcal{I}_2$ at $N = 1, \dots, 5$ are 659, 3474, 10983, 26606, 54675; of $\mathcal{I}_1$, 492, 27032, 263772, 1415016, 5415116.)

Now $I_2(N) > 0$ says $h(N)^2 > 2/\psi(N+1)$, and $R_N > \psi(N+1)$ by (7), so $h(N)^2 > 2/\psi(N+1) > 2/R_N = V^2$ and hence $V < h(N)$. Symmetrically $I_1(N) > 0$ says $\ell(N)^2 < 2/\psi_3(N+1)$, and $R_N < \psi_3(N+1)$, so $\ell(N)^2 < 2/\psi_3(N+1) < 2/R_N = V^2$ and $V > \ell(N)$. Thus

$$(12) \quad (2N+1) + \frac{1}{4N+3} < V < (2N+1) + \frac{1}{4N+2} \quad (N \geq 1).$$

Since $0 < \frac{1}{4N+2} < 1$, (12) gives $\lfloor V \rfloor = 2N+1$, i.e. $a_1(g(N)) = 2N+1$. Taking reciprocals of the fractional part in (12), $1/(V - (2N+1))$ lies strictly between $4N+2$ and $4N+3$, so its floor is $4N+2$, i.e. $a_2(g(N)) = 4N+2$. $\square$

Corollary 5. $a_1(g)|_{N_k} = Q_{2k+1}$ and $a_2(g)|_{N_k} = 2Q_{2k+1}$ for every $k \geq 1$; that is, (2) holds, including at $k = 1$.

Proof. $N_k = (Q_{2k+1} - 1)/2$ gives $2N_k + 1 = Q_{2k+1}$ and $4N_k + 2 = 2Q_{2k+1}$ identically; apply Theorem 1 at $N = N_k \geq 3$. $\square$

4. THE THIRD QUOTIENT

The displayed third quotient is quasi-polynomial of period 19: writing $N = 19s + \rho$ with $0 \leq \rho \leq 18$,

$$(13) \quad A_3(N) := \left\lfloor \frac{12N+6}{19} \right\rfloor = 12s + \left\lfloor \frac{12\rho+6}{19} \right\rfloor.$$

Fix ρ and put, with $A_1 = 2N + 1$ and $A_2 = 4N + 2$ as above,

$$\ell_3 = A_1 + \frac{1}{A_2 + \frac{1}{A_3}}, \quad h_3 = A_1 + \frac{1}{A_2 + \frac{1}{A_3 + 1}},$$

which are rational functions of s with $\ell_3 < h_3$. Because $a_1 = A_1$ and $a_2 = A_2$ are already known from Theorem 1, $a_3(g(N)) = A_3$ holds if and only if $\ell_3 < V < h_3$. (Both ℓ_3 and h_3 require $A_3 \geq 1$; every class below is certified only from an s_0 at which $A_3 \geq 1$, the extreme case being $\rho = 0$, $s_0 = 1$, $A_3 = 12$.) Exactly as in Section 3, using (8) in place of (7), $\ell_3 < V$ follows from the positivity of

$$J_1(s) = 2 - \ell_3^2 \psi_3(N + 1) \Big|_{N=19s+\rho},$$

and $V < h_3$ from the positivity of

$$J_2(s) = h_3^2 \left(\psi_3(N + 1) - \frac{49}{40N^{10}} \right) - 2 \Big|_{N=19s+\rho}.$$

(No separate check that the lower bound in (8) is positive is needed: if it were not, J_2 would be at most -2 .)

Proposition 6. *For each of the 19 residue classes ρ , the numerators and denominators of both J_1 and J_2 , written as polynomials in $s - s_0$, have all coefficients non-negative, with $s_0 = s_0(\rho)$ as tabulated:*

ρ	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\lfloor (12\rho + 6)/19 \rfloor$	0	0	1	2	2	3	4	4	5	6	6	7	7	8	9	9	10	11	11
$s_0(\rho)$	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Consequently $a_3(g(N)) = A_3(N)$ for every $N \geq 5$, and the only integers $N \geq 1$ not covered are $N \in \{1, 2, 3, 4\}$.

Proof. Each entry is a finite exact computation with integer-coefficient polynomials in one variable: substitute $N = 19s + \rho$ and $A_3 = 12s + \lfloor (12\rho + 6)/19 \rfloor$, clear denominators, shift $s \mapsto s_0 + t$ and read the signs. As the covering statement: the classes $\rho = 5, \dots, 18$ certify from $s = 0$, giving $N = 5, \dots, 18$ and then $N = 19s + \rho$ for all $s \geq 1$; the classes $\rho = 0, 1, 2, 3, 4$ certify from $s = 1$, giving $N = 19, 20, 21, 22, 23$ and upwards. The union is $\{N : N \geq 5\}$, and its complement in $\mathbb{Z}_{\geq 1}$ is $\{1, 2, 3, 4\}$. $\square$

In the class $\rho = 9$ one has $12N + 6 \equiv 0 \pmod{19}$, so the floor in (13) sits on an integer; that class too certifies from $s = 0$, with

$$(14) \quad \begin{aligned} \text{num } J_1 \Big|_{\rho=9} &= 12491451 + 142067432s + 671815989s^2 + 1690280508s^3 \\ &\quad + 2385550453s^4 + 1789910922s^5 + 557513238s^6, \end{aligned}$$

of degree 6 with all coefficients positive, while $\text{num } J_2$ has 18 positive coefficients, the constant one being 37139362880053056000. Here num is the numerator over $12(N + 1)^8(A_2A_3 + 1)^2$, evaluated at $N = 19s + 9$ with

$A_2 = 4N + 2$ and $A_3 = 12s + 6$; as a check on the normalisation, the right-hand side of (14) takes the values 12491451, 7249629993 and 147632938319 at $s = 0, 1, 2$.

5. THE FOUR REMAINING INDICES

The four indices $N \in \{1, 2, 3, 4\}$ left over by Proposition 6 are decided by hand from Lemma 4 together with an exact rational partial sum $\sum_{j=N+1}^J j^{-3}$, for one small J . Since $R_N = \sum_{j=N+1}^J j^{-3} + R_J$, applying (7) at J gives

$$(15) \quad \begin{aligned} \frac{155}{768} < R_1 < \frac{207}{1024} & \quad (J = 1), & g(1) &= [0; 3, 6, 1, \dots], \\ \frac{337}{4374} < R_2 < \frac{6067}{78732} & \quad (J = 2), & g(2) &= [0; 5, 10, 1, \dots], \\ \frac{1967}{49152} < R_3 < \frac{10491}{262144} & \quad (J = 3), & g(3) &= [0; 7, 14, 2, \dots], \\ \frac{1707247}{69984000} < R_4 < \frac{61461017}{2519424000} & \quad (J = 5), & g(4) &= [0; 9, 18, 2, \dots]. \end{aligned}$$

In each line the printed bracket already determines the first three partial quotients: one applies $x \mapsto 1/x$ and $x \mapsto x - \lfloor x \rfloor$ three times to the two endpoints and observes that the two integer parts agree at every step. This completes the proof of Theorem 2: the displayed values at $N = 2, 3, 4$ are 1, 2, 2, which (15) confirms, while at $N = 1$ the displayed value is $\lfloor 18/19 \rfloor = 0$ and the true value is 1.

Remark (the exceptional index). $\lfloor (12N + 6)/19 \rfloor = 0$ can only happen at $N = 1$, and 0 is not a legal partial quotient at a position ≥ 1 ; so under the numeric reading the third entry of (1) needs the hypothesis $N \geq 2$. That reading may not be the intended one: (1) is captioned “the *formal* continued fraction”, and under a strictly formal, series-level reading it makes no assertion at the integer $N = 1$.

6. SCOPE

Not settled. The trailing ellipsis of (1). Nothing here addresses a_4, a_5 or their quasi-periods, and the phrase “periods 1, 1, 19, $\dots$ ” therefore remains a conjecture: **the Conjecture of Section 1 is not proved**. Nor is anything proved about $a_i(g(N))$ for $i \geq 4$, about the growth of the quotients, or about irrationality or normality questions for $\zeta(3)$.

No Lean. We wrote no Lean and inspected none: what is discharged is a mathematical statement, not a machine proof.

Attribution. The values $2N + 1$, $4N + 2$ and $\lfloor (12N + 6)/19 \rfloor$ are Feldman’s, printed in [1], which also observes that (2) is immediate from $N_k = (Q_{2k+1} - 1)/2$ once the quotients are known.

Reproduction. Everything above is finite and exact. Lemmas 3 and 4, Theorem 1 and Proposition 6 are polynomial identities and coefficient-sign readings in $\mathbb{Z}[N]$ or $\mathbb{Z}[s]$; Section 5 is four rational brackets. No floating-point number, no enumeration, no solver and no decimal expansion of $\zeta(3)$

enters any proof. The accompanying program `verify.py` re-derives every displayed quantity from the definitions (3)–(4) alone, in exact integer and `Fraction` arithmetic and using only the Python standard library, and reports 95 passing checks; the recorded run `verify.output.txt` also states, in the program’s own words, what it does not cover.

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